

DBT Self Learning

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CDAC Admission Process Case Study

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Windows PowerShell x Windows PowerShell x + v
KD4_98828_Gayatri@localhost>DROP TABLE IF EXISTS students;
Query OK, 0 rows affected (0.206 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>-- 1. TABLE CREATION WITH CASCADE CONSTRAINTS
Query OK, 0 rows affected (0.006 sec)

KD4_98828_Gayatri@localhost>CREATE TABLE students (
-> student_id INT PRIMARY KEY AUTO_INCREMENT,
-> name VARCHAR(100) NOT NULL,
-> email VARCHAR(100) UNIQUE NOT NULL,
-> mobile VARCHAR(15) NOT NULL
-> );
Query OK, 0 rows affected (0.384 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>CREATE TABLE centers (
-> center_id INT PRIMARY KEY AUTO_INCREMENT,
-> center_code VARCHAR(10) UNIQUE NOT NULL,
-> center_name VARCHAR(100) NOT NULL,
-> city VARCHAR(50) NOT NULL
-> );
Query OK, 0 rows affected (0.394 sec)

KD4_98828_Gayatri@localhost>CREATE TABLE courses (
-> course_id INT PRIMARY KEY AUTO_INCREMENT,
-> course_code VARCHAR(10) UNIQUE NOT NULL,
-> course_name VARCHAR(100) NOT NULL
-> );
Query OK, 0 rows affected (0.433 sec)

KD4_98828_Gayatri@localhost>-- Center-Course Capacities
Query OK, 0 rows affected (0.006 sec)

KD4_98828_Gayatri@localhost>CREATE TABLE center_courses (
-> center_course_id INT PRIMARY KEY AUTO_INCREMENT,
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Query OK, 0 rows affected (0.433 sec)

KD4_98828_Gayatri@localhost>-- Center-Course Capacities
Query OK, 0 rows affected (0.006 sec)

KD4_98828_Gayatri@localhost>CREATE TABLE center_courses (
-> center_course_id INT PRIMARY KEY AUTO_INCREMENT,
-> center_id INT NOT NULL,
-> course_id INT NOT NULL,
-> capacity INT NOT NULL,
-> filled_seats INT DEFAULT 0,
-> FOREIGN KEY (center_id) REFERENCES centers(center_id) ON DELETE CASCADE ON UPDATE CASCADE,
-> FOREIGN KEY (course_id) REFERENCES courses(course_id) ON DELETE CASCADE ON UPDATE CASCADE,
-> CONSTRAINT unique_center_course UNIQUE (center_id, course_id)
-> );
Query OK, 0 rows affected (0.723 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>-- Student C-CAT Ranks
Query OK, 0 rows affected (0.005 sec)

KD4_98828_Gayatri@localhost>CREATE TABLE student_ranks (
-> student_id INT PRIMARY KEY,
-> ccat_rank INT NOT NULL UNIQUE,
-> FOREIGN KEY (student_id) REFERENCES students(student_id) ON DELETE CASCADE ON UPDATE CASCADE
-> );
Query OK, 0 rows affected (0.445 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>-- Student Choice Preferences
Query OK, 0 rows affected (0.003 sec)

KD4_98828_Gayatri@localhost>CREATE TABLE preferences (
-> preference_id INT PRIMARY KEY AUTO_INCREMENT,
-> student_id INT NOT NULL,
-> center_course_id INT NOT NULL,
-> preference_number INT NOT NULL,
```

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Query OK, 0 rows affected (0.443 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>-- Final Output Allocations
Query OK, 0 rows affected (0.004 sec)

KD4_98828_Gayatri@localhost>CREATE TABLE allocations (
-> allocation_id INT PRIMARY KEY AUTO_INCREMENT,
-> student_id INT NOT NULL UNIQUE,
-> center_course_id INT NOT NULL,
->
-> allocated_round INT NOT NULL,
-> allocation_status VARCHAR(20) DEFAULT 'ALLOCATED', -- 'ALLOCATED', 'UPGRADED'
-> FOREIGN KEY (student_id) REFERENCES students(student_id) ON DELETE CASCADE ON UPDATE CASCADE,
-> FOREIGN KEY (center_course_id) REFERENCES center_courses(center_course_id) ON DELETE CASCADE ON UPDATE CASCADE
-> );
Query OK, 0 rows affected (0.497 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>-- =====
Query OK, 0 rows affected (0.014 sec)

KD4_98828_Gayatri@localhost>-- 2. SAMPLE RECORD MOCK VALUES
Query OK, 0 rows affected (0.006 sec)

KD4_98828_Gayatri@localhost>-- =====
Query OK, 0 rows affected (0.004 sec)

KD4_98828_Gayatri@localhost>INSERT INTO students (name, email, mobile) VALUES
-> ('Amit Sharma', 'amit@gmail.com', '9876543210'),
-> ('Priya Patel', 'priya@gmail.com', '9876543211'),
-> ('Rahul Verma', 'rahul@gmail.com', '9876543212'),
-> ('Sneha Reddy', 'sneha@gmail.com', '9876543213'),
-> ('Vikram Singh', 'vikram@gmail.com', '9876543214');
Query OK, 5 rows affected (0.183 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

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Windows PowerShell x Windows PowerShell x + v

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>INSERT INTO centers (center_code, center_name, city) VALUES
-> ('IACSD', 'IACSD Akurdi', 'Pune'),
-> ('ACTS', 'CDAC ACTS Pashan', 'Pune'),
-> ('VITA', 'VITA Juhu', 'Mumbai');
Query OK, 3 rows affected (0.050 sec)
Records: 3 Duplicates: 0 Warnings: 0

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>INSERT INTO courses (course_code, course_name) VALUES
-> ('DAC', 'PG Diploma in Advanced Computing'),
-> ('DMC', 'PG Diploma in Mobile Computing');
Query OK, 2 rows affected (0.048 sec)
Records: 2 Duplicates: 0 Warnings: 0

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>-- Map Options and Mock Capacities
Query OK, 0 rows affected (0.003 sec)

KD4_98828_Gayatri@localhost>INSERT INTO center_courses (center_id, course_id, capacity) VALUES
-> (2, 1, 2), -- ID 1: ACTS - DAC (Capacity: 2)
-> (1, 1, 1), -- ID 2: IACSD - DAC (Capacity: 1)
-> (3, 1, 2), -- ID 3: VITA - DAC (Capacity: 2)
-> (2, 2, 1); -- ID 4: ACTS - DMC (Capacity: 1)
Query OK, 4 rows affected (0.043 sec)
Records: 4 Duplicates: 0 Warnings: 0

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>-- Assign Merit Ranks (Lower value means superior rank performance)
Query OK, 0 rows affected (0.002 sec)

KD4_98828_Gayatri@localhost>INSERT INTO student_ranks (student_id, ccat_rank) VALUES
-> (1, 10), (2, 25), (3, 40), (4, 55), (5, 120);
Query OK, 5 rows affected (0.047 sec)
Records: 5 Duplicates: 0 Warnings: 0
```



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Windows PowerShell x Windows PowerShell x + v
KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>DELIMITER ;
KD4_98828_Gayatri@localhost>CALL ProcessRound1Allocation();
+-----+-----+-----+-----+-----+-----+-----+
| allocation_id | name      | ccat_rank | center_code | course_code | allocated_round | allocation_status |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Amit Sharma | 10 | ACTS | DAC | 1 | ALLOCATED |
| 2 | Priya Patel | 25 | ACTS | DAC | 1 | ALLOCATED |
| 3 | Rahul Verma | 40 | IACSD | DAC | 1 | ALLOCATED |
| 4 | Sneha Reddy | 55 | ACTS | DMC | 1 | ALLOCATED |
+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (1.065 sec)

Query OK, 0 rows affected (1.242 sec)

KD4_98828_Gayatri@localhost>DELIMITER //
KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>CREATE PROCEDURE ProcessRound2Allocation()
-- BEGIN
-- DECLARE done INT DEFAULT FALSE;
-- DECLARE current_student_id INT;
-- DECLARE current_rank INT;
-- DECLARE current_allocated_cc_id INT;
-- DECLARE current_allocated_pref_num INT;
-- DECLARE better_pref_cc_id INT;
--
-- DECLARE student_cursor CURSOR FOR
-- SELECT r.student_id, r.ccat_rank FROM student_ranks r ORDER BY r.ccat_rank ASC;
-- DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
--
-- OPEN student_cursor;
-- student_loop: LOOP
--   FETCH student_cursor INTO current_student_id, current_rank;
--   IF done THEN LEAVE student_loop; END IF;
--
--   SET current_allocated_cc_id = NULL;
--
--   SELECT center_course_id INTO current_allocated_cc_id
--   FROM allocations WHERE student_id = current_student_id LIMIT 1;
--
--   IF current_allocated_cc_id IS NOT NULL THEN
--     -- Upgradation Logic
--     SELECT preference_number INTO current_allocated_pref_num FROM preferences
--     WHERE student_id = current_student_id AND center_course_id = current_allocated_cc_id;
--
--     BEGIN
--       DECLARE upgrade_done INT DEFAULT FALSE;
--       DECLARE upgraded INT DEFAULT FALSE;
--       DECLARE upgrade_cursor CURSOR FOR
--       SELECT p.center_course_id FROM preferences p
--       WHERE p.student_id = current_student_id AND p.preference_number < current_allocated_pref_num
--       ORDER BY p.preference_number ASC;
--       DECLARE CONTINUE HANDLER FOR NOT FOUND SET upgrade_done = TRUE;
--
--       OPEN upgrade_cursor;
--       upgrade_loop: LOOP
--         FETCH upgrade_cursor INTO better_pref_cc_id;
--         IF upgrade_done OR upgraded THEN LEAVE upgrade_loop; END IF;
--
--         IF (SELECT (capacity - filled_seats) FROM center_courses WHERE center_course_id = better_pref_cc_id) > 0 THEN
--           UPDATE center_courses SET filled_seats = filled_seats - 1 WHERE center_course_id = better_pref_cc_id;
--           UPDATE allocations SET center_course_id = better_pref_cc_id, allocated_round = 2, allocation_status =
--           'UPGRADED' WHERE student_id = current_student_id;
--           UPDATE center_courses SET filled_seats = filled_seats + 1 WHERE center_course_id = better_pref_cc_id;
--           SET upgraded = TRUE;
--         END IF;
--       END LOOP upgrade_loop;
--     END IF;
--   END IF;
-- END LOOP student_loop;
```

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Windows PowerShell x Windows PowerShell x + v
--
--   FETCH student_cursor INTO current_student_id, current_rank;
--   IF done THEN LEAVE student_loop; END IF;
--
--   SET current_allocated_cc_id = NULL;
--
--   SELECT center_course_id INTO current_allocated_cc_id
--   FROM allocations WHERE student_id = current_student_id LIMIT 1;
--
--   IF current_allocated_cc_id IS NOT NULL THEN
--     -- Upgradation Logic
--     SELECT preference_number INTO current_allocated_pref_num FROM preferences
--     WHERE student_id = current_student_id AND center_course_id = current_allocated_cc_id;
--
--     BEGIN
--       DECLARE upgrade_done INT DEFAULT FALSE;
--       DECLARE upgraded INT DEFAULT FALSE;
--       DECLARE upgrade_cursor CURSOR FOR
--       SELECT p.center_course_id FROM preferences p
--       WHERE p.student_id = current_student_id AND p.preference_number < current_allocated_pref_num
--       ORDER BY p.preference_number ASC;
--       DECLARE CONTINUE HANDLER FOR NOT FOUND SET upgrade_done = TRUE;
--
--       OPEN upgrade_cursor;
--       upgrade_loop: LOOP
--         FETCH upgrade_cursor INTO better_pref_cc_id;
--         IF upgrade_done OR upgraded THEN LEAVE upgrade_loop; END IF;
--
--         IF (SELECT (capacity - filled_seats) FROM center_courses WHERE center_course_id = better_pref_cc_id) > 0 THEN
--           UPDATE center_courses SET filled_seats = filled_seats - 1 WHERE center_course_id = better_pref_cc_id;
--           UPDATE allocations SET center_course_id = better_pref_cc_id, allocated_round = 2, allocation_status =
--           'UPGRADED' WHERE student_id = current_student_id;
--           UPDATE center_courses SET filled_seats = filled_seats + 1 WHERE center_course_id = better_pref_cc_id;
--           SET upgraded = TRUE;
--         END IF;
--       END LOOP upgrade_loop;
--     END IF;
--   END IF;
-- END LOOP student_loop;
```

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Windows PowerShell x Windows PowerShell x + v
--> UPDATE center_courses SET filled_seats = filled_seats - 1 WHERE center_course_id = current_allocated_c
c_id;
--> UPDATE allocations SET center_course_id = better_pref_cc_id, allocated_round = 2, allocation_status =
'UPGRADED' WHERE student_id = current_student_id;
--> UPDATE center_courses SET filled_seats = filled_seats + 1 WHERE center_course_id = better_pref_cc_id;
--> SET upgraded = TRUE;
--> END IF;
--> END LOOP upgrade_loop;
--> CLOSE upgrade_cursor;
--> END;
--> ELSE
--> -- Allocation configuration for previously unallocated students
--> BEGIN
--> DECLARE fresh_done INT DEFAULT FALSE;
--> DECLARE fresh_allocated INT DEFAULT FALSE;
--> DECLARE fresh_cursor CURSOR FOR
--> SELECT p.center_course_id FROM preferences p WHERE p.student_id = current_student_id ORDER BY p.preference
_number ASC;
--> DECLARE CONTINUE HANDLER FOR NOT FOUND SET fresh_done = TRUE;
--> OPEN fresh_cursor;
--> fresh_loop: LOOP
--> FETCH fresh_cursor INTO better_pref_cc_id;
--> IF fresh_done OR fresh_allocated THEN LEAVE fresh_loop; END IF;
--> IF (SELECT (capacity - filled_seats) FROM center_courses WHERE center_course_id = better_pref_cc_id) > 0 T
HEN
--> INSERT INTO allocations (student_id, center_course_id, allocated_round, allocation_status)
--> VALUES (current_student_id, better_pref_cc_id, 2, 'ALLOCATED');
--> UPDATE center_courses SET filled_seats = filled_seats + 1 WHERE center_course_id = better_pref_cc_id;
--> SET fresh_allocated = TRUE;
--> END IF;
--> END LOOP fresh_loop;
--> CLOSE fresh_cursor;
--> END;
--> END IF;
--> END LOOP student_loop;
```

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Windows PowerShell x Windows PowerShell x + v
--> SELECT a.allocation_id, s.name, r.ccat_rank, cen.center_code, c.course_code, a.allocated_round, a.allocation_status
--> FROM allocations a
--> JOIN students s ON a.student_id = s.student_id
--> JOIN student_ranks r ON s.student_id = r.student_id
--> JOIN center_courses cc ON a.center_course_id = cc.center_course_id
--> JOIN centers cen ON cc.center_id = cen.center_id
--> JOIN courses c ON cc.course_id = c.course_id
--> ORDER BY r.ccat_rank ASC;
--> END //
Query OK, 0 rows affected (0.070 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>DELIMITER ;
KD4_98828_Gayatri@localhost>CALL ProcessRound2Allocation();

+-----+-----+-----+-----+-----+-----+-----+
| allocation_id | name       | ccat_rank | center_code | course_code | allocated_round | allocation_status |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Amit Sharma | 10 | ACTS | DAC | 1 | ALLOCATED |
| 2 | Priya Patel | 25 | ACTS | DAC | 1 | ALLOCATED |
| 3 | Rahul Verma | 40 | IACSD | DAC | 1 | ALLOCATED |
| 4 | Sneha Reddy | 55 | ACTS | DMC | 1 | ALLOCATED |
+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.042 sec)

Query OK, 0 rows affected (0.127 sec)

KD4_98828_Gayatri@localhost>
```